IDW interpolation in QGIS

Starting point: the file kleinsee_EM_all_profiles.csv loaded as a point layer, with coordinates and EM values for all four profiles.

Step 1 — Load the CSV as a point layer

If not already loaded: Layer → Add Layer → Add Delimited Text Layer.

1. X field: easting_x, Y field: northing_y
2. CRS: EPSG:32633
3. Add.

Step 2 — Open the IDW interpolation tool

Open the Processing Toolbox (Ctrl+Alt+T) and search for IDW interpolation.

Step 3 — Configure the tool

4. Vector layer: kleinsee_EM_all_profiles
5. Interpolation attribute: EM_HL_mSm (or EM_VL_mSm)
6. Click the + button to add it to the list. Select Puntos as type.
7. Distance coefficient P: leave at 2
8. Extent: click the small button to the right → Calculate from layer → kleinsee_EM_all_profiles
9. Pixel size X and Y: set both to 2 (meters)
10. Run.

A raster layer appears on the map in grayscale. That is the interpolated surface.

Step 4 — Improve the color display

Right-click the interpolated layer → Properties → Symbology.

11. Change Single gray band to Singleband pseudocolor
12. Color ramp: Spectral or RdYlGn reversed
13. Click Classify → OK.

The map now shows conductivity values as a color gradient across the survey area.

Step 5 — Repeat for the other configuration

To visualize EM_VL_mSm as well, run the IDW tool again with the same settings but change the interpolation attribute to EM_VL_mSm. Rename the two raster layers to keep track of which is which.

Notes

IDW interpolates by weighting nearby points more than distant ones. With only four parallel profiles the interpolation between them is an estimate — the result is more reliable along the profile lines than in the gaps between them.

The distance coefficient P controls how quickly the influence of a point drops off with distance. P=2 is the standard value. A higher P gives sharper transitions, a lower P gives a smoother surface.

If the result looks too coarse, reduce the pixel size to 1 m and run again.